

Compact Widely Tunable InP Laser with 10 kHz Linewidth

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Abstract We demonstrate a monolithic InP laser foundry-fabricated on a multi-project wafer that employs two passive parallel ring resonators, targeting coherent interconnects. It achieves 10 kHz intrinsic linewidth, 46 nm tuning range, >35 dB SMSR and 6 mW on-chip output power within a 0.8 mm² footprint. ©2026 The Author(s).

Introduction

The scaling of data center interconnects (DCIs) to higher capacities is driving the adoption of coherent wavelength-division multiplexing (WDM), as conventional intensity-modulation and direct-detection schemes face fundamental power, density, and scalability limitations [1][2][3]. In coherent systems, laser noise directly constrains the achievable modulation format [3]. Narrow-linewidth tunable lasers are therefore essential as both transmitter sources and local oscillators, while compact dimensions are necessary for high integration density.

Significant progress toward narrow-linewidth photonic integrated circuit (PIC) lasers has been achieved through heterogeneous integration of III-V gain materials with low-loss silicon external ring resonators (RR), with intrinsic linewidths as low as 0.14 kHz and 120 nm tuning range [4]. However, monolithic InP integration uniquely enables the straightforward integration of lasers, semiconductor optical amplifiers (SOA), high-speed modulators, and photodetectors within a common material system on a single substrate [5]. This capability enables the incorporation of narrow-linewidth laser sources into complex coherent transmitter PICs [2][6], offering high integration density and reliable large-scale volume production through foundry-compatible generic InP technology.

Despite these advantages, monolithic InP lasers have not yet matched the linewidth performance of their heterogeneous counterparts. Notably, previously demonstrated InP ring laser designs have prioritized fast tunability by employing doped electro-optic modulator waveguides within the cavity, which introduce higher propagation losses and thus fundamentally broaden the

linewidth. For example, in [7] a racetrack configuration employing two parallel-coupled, electro-optically tuned RR achieved a linewidth of ~100 kHz and 34 nm of tuning range. In [8], the advantages of incorporating low-loss passive waveguides in the ring cavity were pointed out, yielding a linewidth of 40 kHz over 43 nm tuning, although the design still retained a short electro-optic phase section. Other monolithic InP approaches include cascaded asymmetric Mach-Zehnder interferometer (AMZI) filters integrated into the laser cavity [9] and extended-cavity distributed Bragg reflector (DBR) lasers incorporating a third grating [10]. A comparison of these prior demonstrations (Tab. 1) shows that combining narrow linewidth, wide tunability, adequate output power levels and a compact footprint within an InP laser architecture remains an open challenge.

In this contribution, we demonstrate a significant improvement in both intrinsic linewidth and chip area for foundry-fabricated monolithic InP ring lasers.

Chip Design

Fig. 1 shows a photograph of a fabricated ring laser chip. The design employs two parallel-coupled ring resonators in a racetrack formation, drawing on the architectures reported in [11][7] and [8], with exclusively low-loss passive waveguides within the laser cavity.

The ring resonators are realized using 1×2 multimode interference couplers (MMI) with a 50/50 power splitting ratio and Euler clothoid bends. Couplers based on symmetric interference are advantageous owing to their broad power-transmission bandwidth. The employed 1×2 MMIs exhibit a 0.5 dB power-transmission

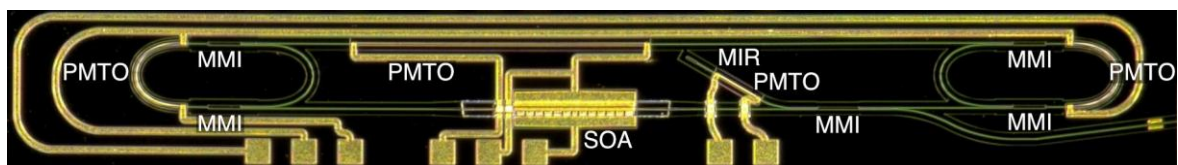


Fig. 1: Fabricated ring laser die with labelled building blocks: semiconductor optical amplifier (SOA), thermo-optic phase modulator (PMTO), multimode interference couplers (MMI), multimode interference reflector (MIR).

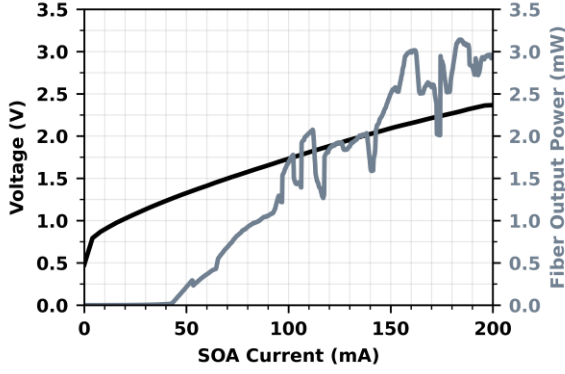


Fig. 2: Measured light-current-voltage characteristics.

bandwidth of approximately 150 nm centered around 1550 nm, with a minimum insertion loss of 0.3 dB. The ring circumferences are $L_1 = 1380 \mu\text{m}$ and $L_2 = 1388 \mu\text{m}$. Exploiting the Vernier effect, the resulting combined free spectral range (FSR) of the dual-ring filter extends to approximately 80 nm. The SOA and an output coupling section are integrated into the waveguides connecting the two parallel-coupled add-drop ring resonators. A distinctive feature of the architecture is that the outer halves of both ring resonators, together with the connecting waveguide sections, form a third "outer" ring cavity. A multimode interference reflector (MIR) is integrated at the output section to enforce unidirectional operation by coupling the clock- and counterclockwise propagating modes.

The device was realized on the generic InP foundry platform of the Fraunhofer Heinrich-Hertz-Institute through an open-access multi-project wafer (MPW) run. Deep-etched passive ridge waveguides with propagation losses below 2 dB/cm are used throughout the ring cavity. Thermo-optic phase shifters (PMTO), implemented using a platinum heater layer sandwiched between two SiN isolation layers, are deposited on top of the passive waveguides for wavelength tuning. The SOA features a multiple quantum well (MQW) active region with a photoluminescence (PL) peak at 1530 nm, situated on

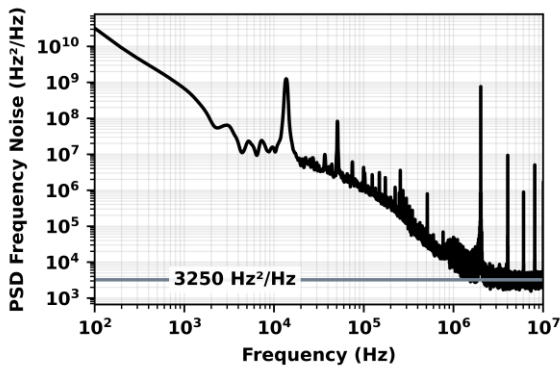


Fig. 4: Measured frequency noise spectrum (RBW = 1 kHz).

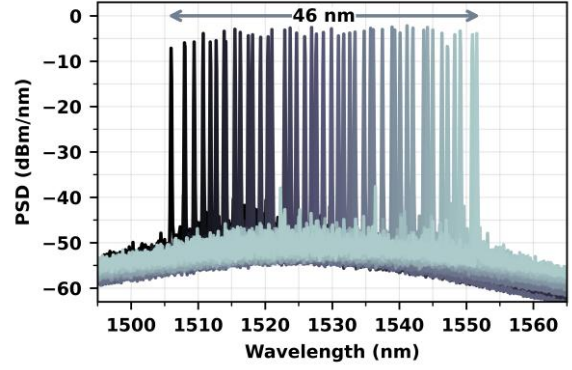


Fig. 3: Superimposed spectra (RBW = 0.1 nm).

a separate active mesa connected to the passive circuit by butt-joint active/passive transitions, each contributing ~ 1 dB of insertion loss. All components used in the fabricated devices are drawn exclusively from standardized building blocks available in the foundry process design kit (PDK).

Device Performance

The fabricated ring laser chips were mounted on a gold-coated copper heat sink and temperature-stabilized at 20 °C using a Peltier element for all measurements. Electrical biasing was provided by PICConnect multi-channel PIC characterization system [12]. To suppress parasitic optical feedback into the cavity, the chip output facet was coated with an anti-reflection (AR) coating. Free-space optical coupling was implemented using an AR-coated lensed fiber, immediately followed by a fiber-coupled optical isolator.

Fig. 2 displays the fiber-coupled output power and SOA voltage as a function of injection current with the laser operating in continuous-wave mode. The fiber-coupled output power reaches 3 mW at 200 mA. The corresponding on-chip power is approximately 6 mW when accounting for 3 dB coupling losses. If required, they can be reduced to 1.5 dB by incorporating a spot-size converter (SSC) available in the PDK. The laser exhibits a threshold current of 44 mA. At injection currents above 50 mA, multiple discontinuities in

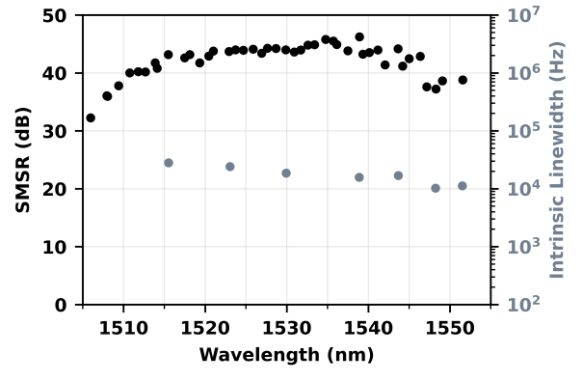


Fig. 5: Wavelength dependence of SMSR and linewidth.

Tab. 1: Performance comparison of narrow-linewidth architectures. Note: The output powers are reported as on-chip values unless marked with *, indicating fiber-coupled output power including coupling losses, or **, indicating a booster SOA at the output.

	Platform	Linewidth in kHz	Tuning Range in nm	SMSR in dB	Power in mW	Chip Area in mm ²
This work	InP	10	46	35	6	0.8
Two parallel RR Loop Mirror [8]	InP	40	43	36	6 *	3.0
Two parallel RR MIR [7]	InP	100	34	50	1.8*	1.2
External cavity DBR [10]	InP	10	0.3	54	18	1.5
Cascaded AMZI [9]	InP	16	51	40	0.9*	3.3
Four-section DBR [2]	InP	200	40	48	40 **	-
TDA Chirped SG-DBR [6]	InP	50	40	45	63 **	-
DFB Array [14]	InP	543	9.6	50	100 **	-
Two parallel coupled RR [11]	Si/InP	338	40	35	3.3	-
Dual ring mirror [4]	Si/InP	3.5	40	50	10	-
Quad ring mirror [4]	Si/InP	0.14	120	40	2	-
Extended DBR [15]	Si/InP	1.1	-	55	37	-

the output power characteristic are observed, linked to mode-hopping behavior caused by the refractive index change in the SOA section detuning the cavity modes.

Increasing the heater current on ring resonator 1 (on the right-hand side in Fig. 1) pre-dominantly shifts the lasing wavelength to longer wavelengths, while increasing the heater current on ring resonator 2 shifts the wavelength to shorter values. The outer ring section phase shifter enables fine adjustment of the SMSR by optimizing the alignment between the Vernier filter peaks of the two ring resonators and the outer cavity resonances. The overall tuning behavior is analogous to that of sampled-grating (SG-) DBR lasers [13]. A total tuning range of 46 nm, from 1506 to 1552 nm, has been obtained. The superimposed spectra at selected operating points across the tuning range are shown in Fig. 3. As is characteristic of laser architectures employing the Vernier effect, the tuning is discrete rather than continuous.

The spectral purity of the laser was characterized through frequency noise measurements using a frequency-discriminator-based optical frequency noise analyzer. The frequency noise spectrum, obtained at 1548 nm, is presented in Fig. 4. In the region from 1 kHz to 1 MHz, the frequency noise exhibits a roll-off attributed to extrinsic technical noise arising from drive-current noise and environmental fluctuations. Above 1 MHz, a frequency-independent white noise floor is estimated at 3250 Hz²/Hz corresponding to an intrinsic Lorentzian linewidth of approximately 10 kHz.

Fig. 5 presents the intrinsic linewidth and the SMSR as a function of emission wavelength across the tuning range. The linewidth remains below 28 kHz for all wavelengths above 1515 nm. The increase in linewidth towards

shorter wavelengths is attributed to the combined effects of the asymmetric gain spectrum and wavelength-dependent intracavity losses. The SMSR reaches a peak of 46 dB at 1538 nm and exceeds 40 dB between 1510 and 1542 nm.

Conclusion and Outlook

In this work, we have demonstrated a compact monolithic InP ring laser achieving an intrinsic Lorentzian linewidth of 10 kHz while maintaining a 46 nm tuning range, >35 dB SMSR, and 6 mW on-chip output power within a footprint of only 0.8 mm².

Tab. 1 provides a direct comparison of narrow-linewidth lasers across different platforms. To our knowledge, this laser currently represents the narrowest-linewidth, widely tunable on-chip light source available on an InP foundry platform. Its compact footprint makes this device a viable narrow-linewidth alternative to DBR and SG-DBR lasers, previously employed in monolithic InP transmitter PICs [2][6]. The difference in output power can be addressed by using a booster SOA at the ring laser output. The tuning range can be adjusted to cover the C-band by shifting the PL peak of the gain medium to longer wavelengths. Furthermore, the linewidth can likely be further reduced to the low-kHz regime by employing larger ring circumferences or lower output coupling ratios.

These results indicate that monolithic InP ring lasers are beginning to close the performance gap with heterogeneous Si/InP solutions. This progress supports the future integration of narrow-linewidth monolithic InP lasers into coherent interconnects.

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